

Mario Alberto Santos Ervedosa

Actuary (AIA) | Credit Risk Specialist
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Portfolio · GitHub · LinkedIn

EXPERIENCE

Credit Risk Modelling Manager, Gini Mar 2026 - Present

- I led a small team validating a Pillar 2A credit concentration risk capital model to a supervisory submission deadline.
- I initiated and built the firm's internal AI capability and led it to firm-wide adoption: over 50 reusable agent skills with automated guardrails and eval suites, and a research knowledge base of over 1,500 graded credit risk and regulatory sources.

Credit Risk Modelling Manager, Deloitte UK

Apr 2025 - Feb 2026

- I supported development and regulatory documentation of PD, LGD and EAD models for unsecured portfolios, ensuring PRA and ECB compliance and strengthening governance for audit readiness.
- I designed a Python-based credit risk code library in Azure, standardising IFRS 9 and IRB modelling workflows and holding reproducibility across teams.

Data Science Consultant

Oct 2024 - Apr 2025

Senior Actuarial Associate, PwC South Africa

Jan 2023 - Sep 2024

- I co-developed, deployed and maintained a first-of-its-kind Strategic Workforce Planning application hosted in Azure. I designed the database and injection process in PostgreSQL, and modelled workforce supply and demand using survival analysis in Python.

Actuarial Associate, PwC South Africa Jan 2021 - Dec 2022

- I validated an XGBoost PD model and built a Power BI dashboard to visualise portfolio trends and outputs for IFRS 9 reporting.
- I performed independent LGD modelling utilising SQL, Python and SAS, while project managing and leading junior team members.
- I supported the testing of a PD model built using a Cox proportional hazard approach, and calibrated an EAD model with limit increase effects.

Professional Swimmer

Jan 2014 - Dec 2020

- I competed internationally at two FINA World Championships.

PROJECTS

Ad campaign modelling

Fount | 2025

I developed and validated GLMs on Facebook/Google marketing data for US insurers to evaluate performance across regional and demographic dimensions.

Crop yield modelling

Agnify | 2024

I developed time series forecasts to predict crop production trends, enabling data-driven planning at farmer, state, and national levels.

SUMMARY

I'm an actuary specialising in credit risk, with six years building and validating the models banks hold capital against, across IRB, IFRS 9 and Pillar 2. I work as readily on a regulatory framework as on the statistics underneath it and the code that produces the numbers. Alongside delivery, I build the tooling that keeps it consistent: shared model libraries, automated guardrails and a graded research base now used across the firm. My sectors span banking, insurance, the public sector and agriculture. In my spare time I play tennis, explore restaurants, go to Catholic seminars and watch sports.

CERTIFICATIONS

Chartered Actuary (Associate) IFoA
Actuary (AMASSA) ASSA
Deep Learning for Actuarial Modelling Università Cattolica
Data Scientist Codecademy
Data Engineer Codecademy

TECH STACK

Languages & Tools

Python | SQL | Power BI | Excel | SAS | Git | Claude

Libraries

scikit-learn | pyspark | duckdb | TensorFlow | shap | pandera | Seaborn | Statsmodels | polars

EDUCATION

BScHons Actuarial Science,
University of Pretoria
Jan 2020 - Dec 2020

BSc Actuarial & Financial
Mathematics (Cum Laude),
University of Pretoria
Jan 2017 - Dec 2019